Solutions–In-Class Exercise

ECON 300 – Labour Economics

Winter 2026

Common Setup (Reminders)

Total time: $T = 80$. Leisure L on horizontal axis, consumption C on vertical axis.
Hours worked: $h = 80 - L$. Wage: $w = 20$.
Baseline (no demogrant):

$$C = 20h = 20(80 - L) = 1600 - 20L.$$

So the baseline budget line:

- passes through $(L = 80, C = 0)$ (no work),
- passes through $(L = 0, C = 1600)$ (work 80 hours),
- has slope $-w = -20$.

Key labour-supply logic (what to look for in answers)

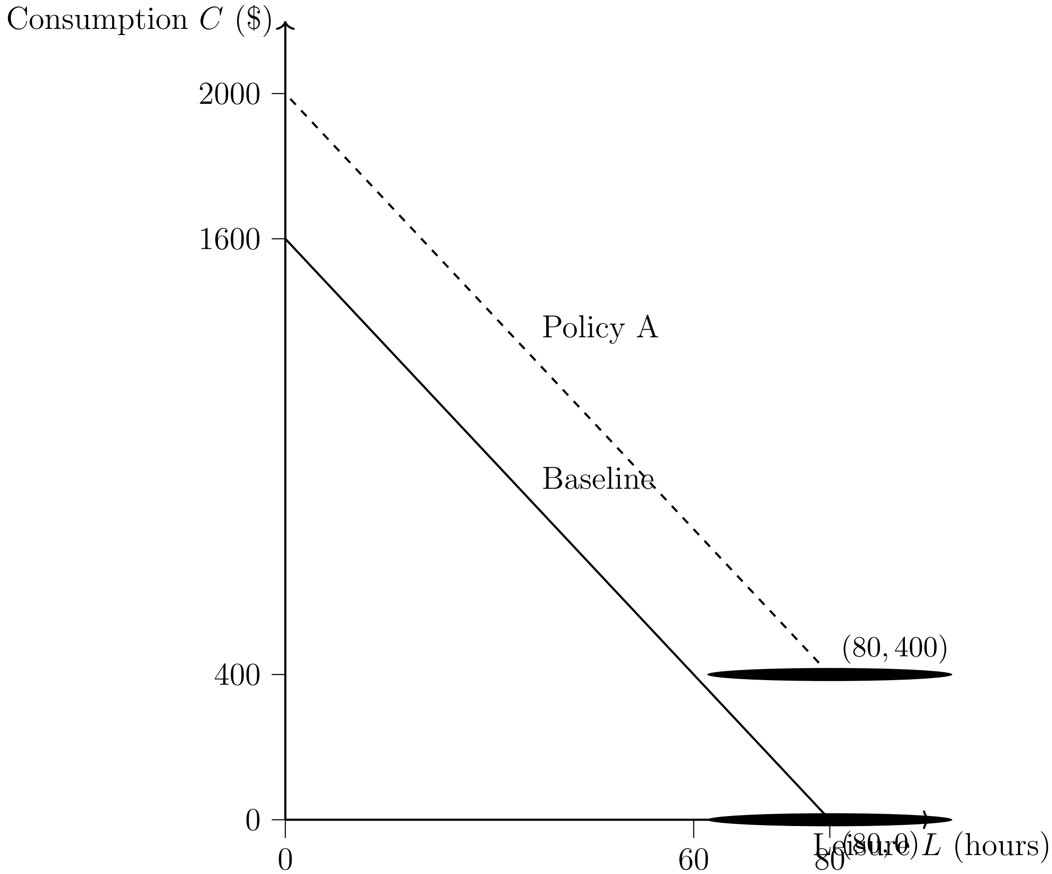
- **Income effect:** More non-labour income (or higher transfer) raises feasible consumption; if leisure is normal, tends to $\uparrow L$ (less work).
- **Substitution effect:** Any policy that reduces the effective net wage (flatter budget line) makes leisure cheaper relative to consumption; tends to $\uparrow L$ (less work). If net wage increases, substitution effect tends to $\downarrow L$ (more work).
- **Participation (extensive margin):** Policies that raise consumption specifically at $h = 0$ or create large kinks/jumps around $h = 0$ often have the largest participation response.
- **Reservation wage:** The minimum wage needed to make working (even a little) preferable to not working. Policies that make $h = 0$ more attractive tend to $\uparrow w^R$.

Policy A: Universal Lump-Sum Demogrant

Policy: Everyone receives \$400 regardless of work.

$$C = 400 + 20h = 400 + 1600 - 20L = 2000 - 20L.$$

Diagram (Policy A vs Baseline)



Answer Key (Policy A)

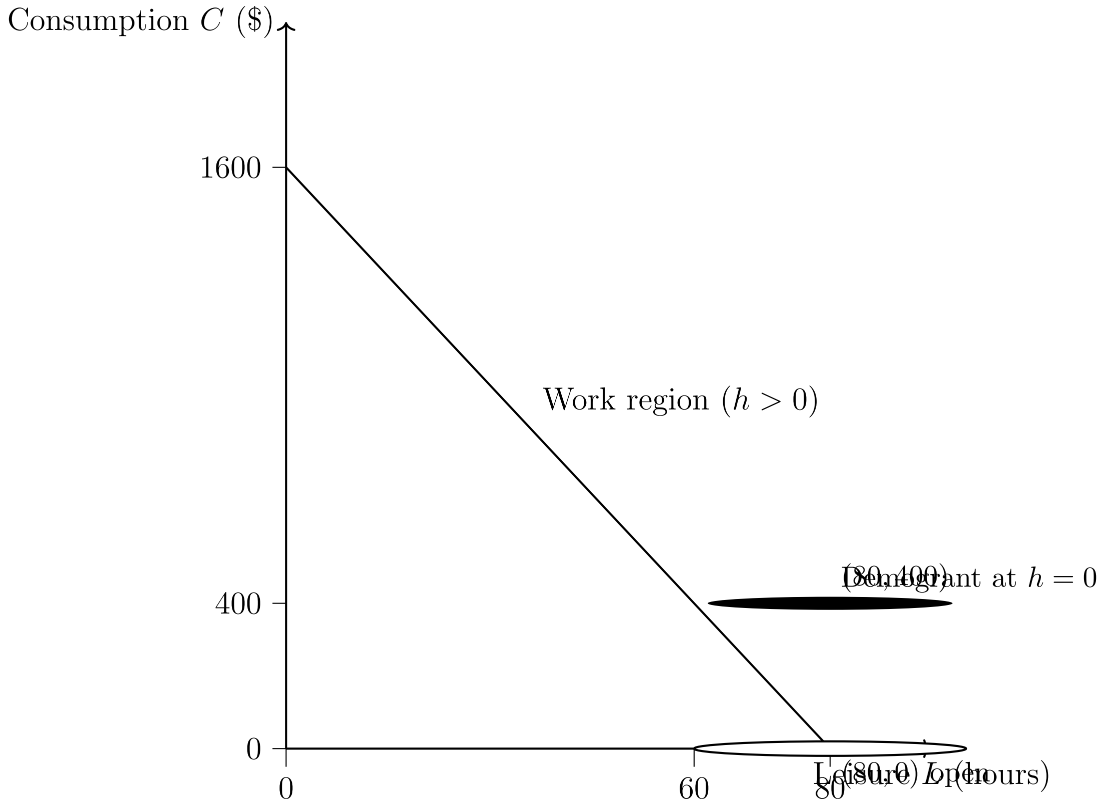
1. **Budget constraint change:** A **parallel upward shift** by \$400. Same slope -20.
2. **Substitution effect? No.** The slope (net wage) is unchanged.
3. **Income effect: Yes.** Higher non-labour income makes the person richer; if leisure is normal, they choose $\uparrow L$ (less work).
4. **Predictions:**
   - **Hours worked:** typically $\downarrow h$ (pure income effect).
   - **Participation:** may fall for marginal participants (those close to indifference at $h = 0$), but generally the strongest effect is on *hours*, not necessarily a large discrete participation jump.

Policy B: Targeted Demogrant (Non-Workers Only)

Policy: \$400 only if $h = 0$.

$$C = \begin{cases} 400 & \text{if } h = 0 \ (L = 80), \\ 20h = 1600 - 20L & \text{if } h > 0 \ (L < 80). \end{cases}$$

Diagram (discontinuity at $h = 0$ / $L = 80$)



Answer Key (Policy B)

1. **Budget set shape:** Same baseline line for $h > 0$, plus a **jump up** to a higher point at $h = 0$ (a discontinuity at $L = 80$).

2. **Reservation wage: Increases.** The outside option ($h = 0$) is now better, so a higher wage is needed to induce participation for marginal workers.

3. **Which margin is most affected? Participation (extensive margin).** The policy directly rewards non-work, creating a strong incentive to choose $h = 0$.

4. **Prediction:** More people choose $h = 0$ (non-participation), especially those with high value of leisure or low market wages.

Instructor note (common misconception): Students may call this a “pure income effect.” It is *not* just income: the design creates a discontinuity that changes the work/non-work choice sharply (a strong extensive-margin distortion).

Policy C: Phased-Out Demogrant (Negative Income Tax Style)

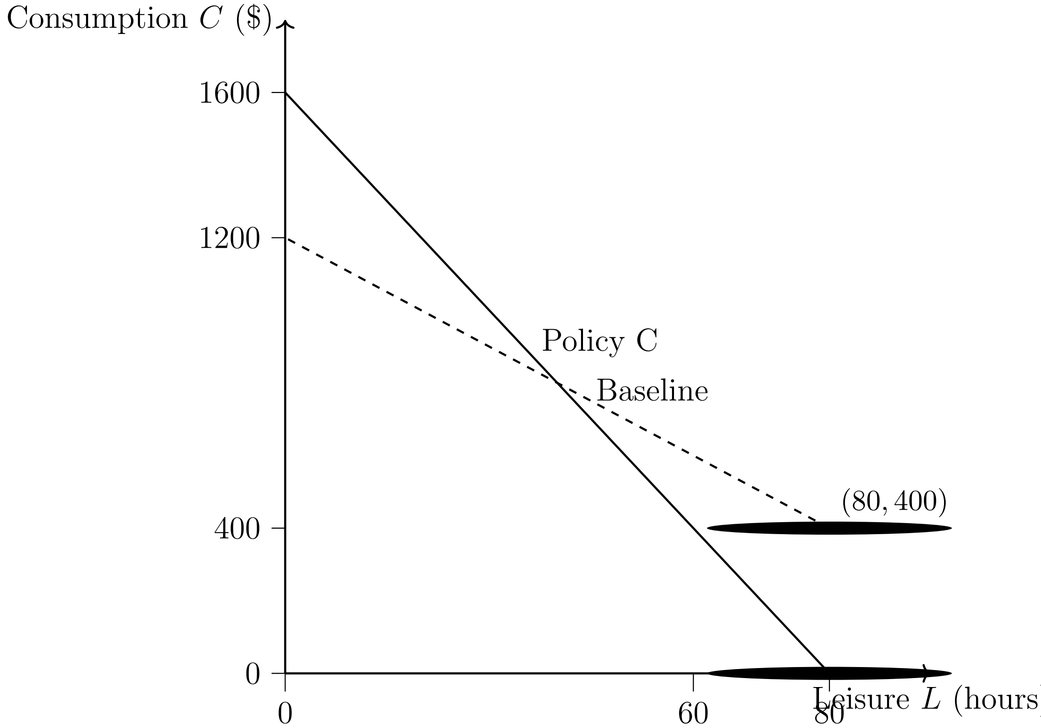
Policy: \$400, phased out at 50% of earnings. Net-of-benefit-reduction wage is lower.

$$C = 400 + (1 - 0.5) \cdot 20h = 400 + 10h.$$

In terms of leisure:

$$C = 400 + 10(80 - L) = 1200 - 10L.$$

Diagram (flatter budget line: lower net wage)



Answer Key (Policy C)

1. **Slope comparison:** Baseline slope is -20. Policy C slope is -10 (flatter). This reflects a lower **net wage** due to benefit phase-out.

2. **Income effect: Yes** (transfer increases non-labour income at any given h, especially near $h = 0$).

3. **Substitution effect: Yes** because the net wage falls from 20 to 10; leisure becomes cheaper relative to consumption, encouraging $\uparrow L$ (less work).

4. **Predictions:**

- **Hours conditional on working:** tends to $\downarrow h$ (both income and substitution reduce work).
- **Participation:** also tends to fall for marginal workers (better outside option + weaker reward to working an extra hour).

Instructor note: This is the classic case where you should expect stronger labour supply reductions than Policy A because Policy A has no substitution effect.

Policy D: Conditional Demogrant (Work Requirement)

Policy: \$400 only if $h \geq 20$ (equivalently $L \leq 60$).

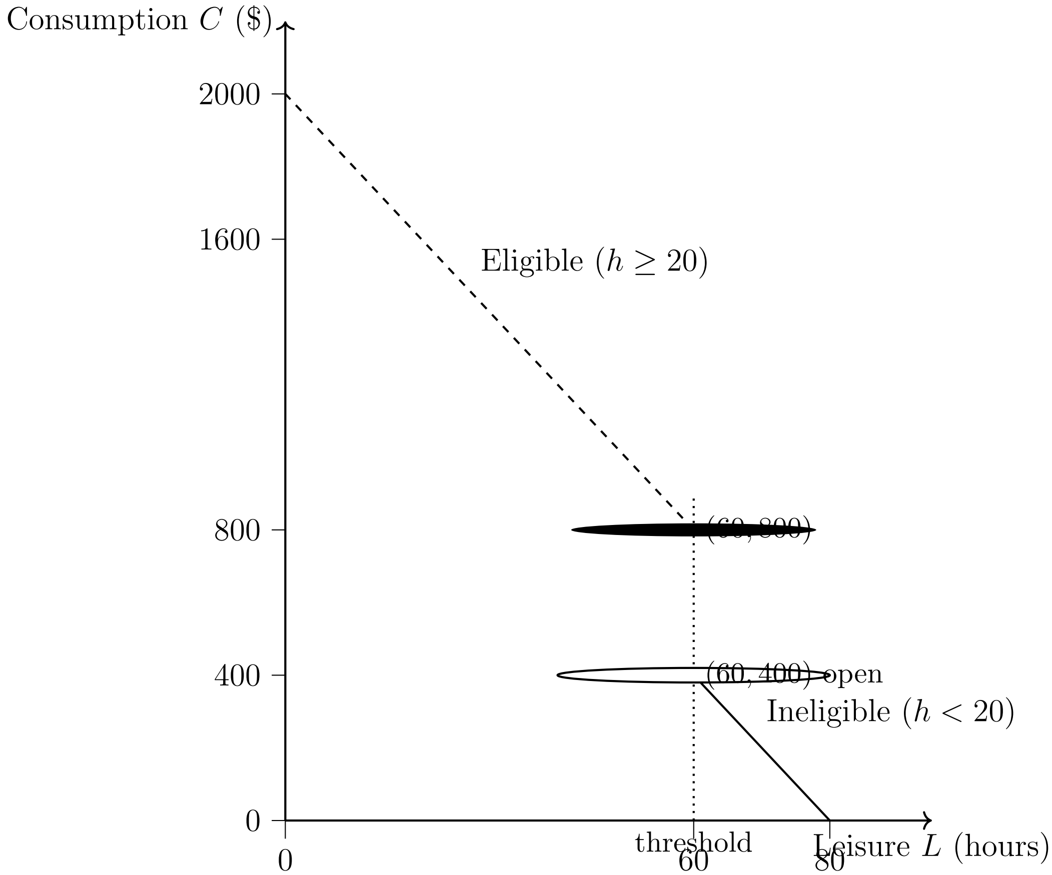
$$C = \begin{cases} 20h = 1600 - 20L & \text{if } h < 20 \ (L > 60), \\ 400 + 20h = 2000 - 20L & \text{if } h \geq 20 \ (L \leq 60). \end{cases}$$

At the threshold $h = 20$ ($L = 60$):

$$C_{\text{below}} = 20 \cdot 20 = 400, \qquad C_{\text{eligible}} = 400 + 20 \cdot 20 = 800.$$

So there is a **jump of \$400** at $L = 60$.

Diagram (jump at eligibility threshold $h = 20$ / $L = 60$)



Answer Key (Policy D)

1. **Budget set shape:** Baseline applies for $h < 20$. At $h = 20$ there is a **discrete jump up** in consumption, then the eligible region is a parallel upward shift (same slope -20).
2. **Effect near 20 hours:** Strong incentive to work at least 20 hours to "qualify" for the lump-sum payment.

3. **Distortion created:** A **notch** / eligibility discontinuity at $h = 20$ (or $L = 60$), which can cause **bunching** (clustering) just at/above 20 hours.

4. **Predictions:**

   - Some individuals who would have chosen slightly less than 20 hours may increase hours to reach eligibility (participation may increase for some, or hours may jump).
   - Individuals already above 20 hours experience a pure income effect (may reduce hours slightly while staying eligible).
   - Overall, expect **clustering at the threshold** and non-smooth labour supply responses.

Wrap-Up Discussion: Suggested Answers

1. **Only income effect:** Policy A (universal lump sum, no change in slope).
   Income + substitution effects: Policy C (phase-out reduces net wage, flattening budget line).
   Discontinuity-driven extensive-margin distortion: Policy B and Policy D.

2. **Most likely to reduce participation:** Policy B (pays only if not working).
   Most likely to reduce hours among workers: Policy C (lower net wage + transfer).
   Policy A tends to reduce hours via income effect; participation effect depends on how close people are to the margin.

3. **Why governments use conditional/phased-out designs:** targeting fiscal resources to low-income households, political preferences for "work incentives," and budget constraints; but these designs often create stronger behavioural distortions (kinks/notches, reduced net wage).